



# Ali Alouane

Machine Learning Engineer

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## About Me

3+ years of professional experience in machine learning and software development. Passionate about artificial intelligence and robotics, with strong focus on applied ML and scalable solutions.

## Social Network

- Ali Alouane
- mralioo
- portfolio-ali-alouane

## Languages

- German
- English
- French
- Arabic

## Expertise

- Machine Learning
- Deep Learning
- Data engineering
- Sensor Fusion
- CI/CD
- MLOps
- Docker
- AWS

## Working Experience

- 11.2024 – **Machine Learning Engineer** ITUC GMBH
  - Designing and implementing scalable ML pipelines for Large Language Models (LLMs) across various business applications.
  - Rapid prototyping of AI-driven solutions, enabling fast iteration cycles and proof-of-concepts.
  - Developing deployment workflows with CI/CD integration to ensure production-readiness and reliability.
  - Conducting systematic model evaluation, benchmarking, and optimization to guarantee performance and robustness.
  - Collaborating with cross-functional teams to translate business concepts into production systems.

**Technologies Used:** LangChain, Github Action, Airflow, MLFlow, FastAPI
- 10.2023 – **Technical Co-Founder & Lead ML Engineer** ROVEONE 08.2024
  - Co-founded an edge-AI civilian protection startup, served on its board until August 2026, and led development of its privacy-first iOS threat detection engine.
  - Designed a Bayesian sensor-fusion engine combining audio, motion, and contextual signals; deployed optimized YAM-Net/CoreML models on the Apple Neural Engine.
  - Built a GDPR-compliant GCP data and active-learning pipeline for sensor ingestion, labeling, model improvement, and production analytics.

**Technologies Used:** TensorFlow, CoreML, Python, Swift, GCP, Docker
- 04.2022 – **ML Engineer – Intern, & Freelancer** TRUEMETRICS GMBH (YC S23) 10.2023
  - Progressed from intern to the startup’s first ML Engineer and continued as a freelancer, taking end-to-end ownership of machine learning development.
  - Analyzed and visualized large-scale mobility and sensor data to uncover actionable insights for logistics applications.
  - Developed, fine-tuned, and evaluated time-series machine learning models for reliable real-world predictions.
  - Built and deployed a production ML pipeline on AWS, covering data processing, feature extraction, experiment tracking, and model inference.

**Technologies Used:** Python, TensorFlow, tsfresh, AWS, MLflow, Scrum
- 09.2018 – **Working Student, Transmission & Hybrid Systems** IAV GMBH 10.2020
  - Developed a Reinforcement Learning framework for automotive applications.
  - Automated data preprocessing pipelines for enhanced efficiency.
  - Implemented and evaluated reinforcement and supervised learning algorithms.

**Technologies Used:** Python, Matlab, PyTorch, jenkins

Hard Skills

-  Deep Learning Techniques
-  IoT
-  MLOps

Soft Skills

-  Critical Thinking
-  Troubleshooting
-  Continuous Learning
-  Teamwork
-  Collaboration
-  Communication
-  Good Listener

Software Skills

-  Python
-  C++
-  DL Frameworks (TensorFlow, PyTorch)
- Docker
-  Git / DVC
- AWS Services

Education

Study

- 2018–2024

M.Sc. in Electrical Engineering

Technical University of Berlin

Specializing in communication networks and signal processing.

Master Thesis

Neurotechnology Group

Deep Learning Approaches to Decoding Sensorimotor Rhythms for Brain-Computer Interfaces.
- 2013–2017

B.Sc. of Electrical Engineering and Information Technology

University of Stuttgart

Specializing in communication networks and signal processing.

Bachelor Thesis

Institute for Signal Processing

Multidimensional energy disaggregation in residential buildings via deep learning.

Projects

- 10.2020

Bearing Fault Diagnosis using Deep Learning

Measurement and Diagnostic Institute

Implemented a DL-based system for classification of bearing damages, enhancing industrial maintenance.
- 07.2020

GNU-Radio Meets OpenAI-Gym

Institute for Telecommunications Systems

Created a framework integrating GNU-Radio with OpenAI Gym for RL agents in SDR scenarios.

Certificates

- Summer 2026

10-Week Startup Bootcamp

Berlin Startup School

Validated a healthcare business idea using structured customer discovery and startup frameworks; learned from experienced entrepreneurs how to test assumptions, develop an MVP, and build a viable business model.
- 08/2024

Deep Learning Institute

NVIDIA

Certifications in Fundamentals of Deep Learning and Accelerated Computing with CUDA Python.
- 05/2024

Design Thinking Intensive Training

.msg

Trained in user research, problem definition, prototyping, and user testing.
- 08/2018

Deep Learning Specialization

deeplearning.ai

Completed Coursera specialization in neural networks, ML, and AI applications.

Extra-Curricular Activities

- DIY Projects

Built projects like an Autonomous Robocar, Smart Home Assistant, and 3D printing models.
- Hackathons

Participant in Climathon, DeepBerlin, and Bosch IoT Hackathon.
- Clubs

Member of Tunisian Space Association and Deep Berlin Robocars.
- Sports

Running and Karate enthusiast.